SWAGNIK ROYCHOUDHURY

Pittsburgh, PA · sroychou@andrew.cmu.edu · swagnik.org · github.com/GOTWIC · linkedin.com/in/swagnik · Google Scholar

EDUCATION

Carnegie Mellon University, Robotics Institute, Master of Science in Robotics (MSR) *(Pittsburgh, PA)* 2026 to 2028

- Advised and funded by Jiaoyang Li on a Toyota Research Institute (TRI) project in skill-based multi-agent long-horizon task planning for manipulation.

New York University, Double Major in Computer Science and Data Science *(New York, NY)* 2022 to 2025

- GPA 3.9, Magna Cum Laude, Dean's List all three years. Funded researcher at NYU Ensure Group.

EXPERIENCE

Founding Engineer, Stealth Robotics Startup *(San Francisco, CA)* 11/2024 to 8/2026

- Built a shared autonomy system for dual-arm manipulation that fuses teleoperator actions with VLA policies, via finetuning on teleop demonstrations and test-time guidance of action sampling from the operator's recent trajectory.
- Built the teleop and policy infrastructure: Unitree G1 and dual-arm teleop and locomotion, MuJoCo simulation, Apple Vision Pro streaming, camera pipelines, and LeRobot dataset tooling. Details under NDA.
- Designed a sensorized UMI gripper variant (force/torque sensors, FSRs) for close-contact tasks with humans, plus its multi-sensor data collection stack and operator dashboard.
- Added a runtime safety layer that bounds the policy's end-effector target around the operator's pose, enforces joint position and velocity limits, and hands off authority between policy and operator without discontinuities.
- Backed by the founders and CEOs of Sherpalo Ventures, Lime, Notion, and Vicarious, among others.

Human-Robot Collaboration Researcher, Montclair State University *(Montclair, NJ)* 6/2024 to 12/2024

- Developed DISCERN, a training-free task ordering pipeline for human-robot collaboration that combines commonsense knowledge, linguistic networks, and a vision language model. Works with about 20,000 ImageNet-21k classes.

Brain-MRI Student Researcher, NYU Center for Data Science *(New York, NY)* 9/2024 to 12/2024

- Developed masked autoencoders and vision transformers on 3D MRI scans and clinical survey data to predict downstream neuropsychiatric symptoms of Alzheimer's.

ICMLM: Language Model for Indian Classical Music, NYU Ensure Lab *(New York, NY)* 8/2023 to 2/2024

- Trained transformers to generate Indian classical music on a first-of-its-kind hand-built dataset, benchmarked against in-context learning and finetuning of GPT-4, Claude 2, and LLaMA 2.

Competition Team Lead, NYU RoboMaster Robotics Team *(New York, NY)* 9/2022 to 5/2024

- Led an 11-member team building autonomous robots: simulation, CV for autonomy, robot CAD, and embedded testing.

Software Developer and Data Science Intern, INVIDI Technologies 6/2022 to 1/2023

- Built ETL for ad impression data and trained RNNs for time series forecasting on AWS (Redshift, SageMaker, S3).

PUBLICATIONS

- DISCERN for Generalized Robotics Contexts. *IEEE Big Data 2024 (poster, full proceedings)*.
- The DISCERN Approach for Intelligent and Efficient Discernment of Robotic Task Contexts. *MIT URTC 2024, IEEE Xplore*.
- Applications of BadNets in Spam Filters. *ICDE 2023 ASTRIDE Workshop*.
- S^2: Information-Theoretically Secure and Highly Efficient Search and Row Retrieval. *VLDB 2023*.
- ICMLM: A Language Model for Indian Classical Music. *NYU Undergraduate Research Conference 2024 (poster)*.
- OccuV: A Visualizer for Spatial and Temporal Databases. *UC Irvine Undergraduate Research Symposium 2023 (poster)*.

AWARDS

- **6x DURF Research Grant Recipient**, NYU (2023 to 2024), including two conference travel grants.
- **Goldwater Scholarship Nominee**, NYU (2023): one of four students nominated.

SKILLS

Languages: Python, C++, C#, SQL, Swift

Robot Learning: VLA and diffusion policy finetuning (GR00T, $\pi_{0.5}$, MoDE), flow matching, imitation learning, reinforcement learning (PPO) for dexterous manipulation, LeRobot

Robotics: MuJoCo, Isaac Sim, OmniGibson, inverse kinematics, VR teleoperation (Apple Vision Pro), force/torque sensing, Unitree G1, YAM, Franka Panda

ML: PyTorch, PyTorch Lightning, CUDA, Hydra, Weights & Biases, transformers, OpenCV

Tools: AWS (EC2, S3, SageMaker), Docker, Linux, Git, MySQL, Unity, Blender, SolidWorks, Xcode